

SolarLink Infrastructure Plan

DevOps & Cloud Engineering — Week 1

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Project: SolarLink Platform Architecture

1. Executive Summary

The SolarLink platform requires a resilient and scalable infrastructure to support its marketplace and monitoring services. The chosen architecture leverages Amazon Web Services (AWS) to provide a cloud-native, containerized environment that ensures high availability and security for users in Nigeria and beyond.

2. Technical Architecture & Services

- **Compute (AWS ECS/Fargate):** Deployment of Docker containers in a serverless environment to eliminate server management overhead.
- **Database (Amazon RDS PostgreSQL):** Managed relational database service for structured data with automated backups and encryption.
- **Storage (Amazon S3):** Secure, durable object storage for solar panel media and user documents.
- **CDN (Amazon CloudFront):** Edge caching to ensure low-latency content delivery across diverse geographical locations.

3. Scalability & Cost Justification

By utilizing serverless compute (Fargate) and managed databases (RDS), SolarLink adopts a "Pay-as-you-go" model. This minimizes initial capital

expenditure while allowing the platform to scale horizontally during peak demand. The separation of storage (S3) and compute ensures that data growth does not negatively impact application performance.

4. Connectivity & Security

The architecture utilizes an Application Load Balancer (ALB) to distribute traffic across containers. Security is maintained through VPC isolation and AWS Secrets Manager for sensitive environment variables, building upon the local environment variable strategy established during the development phase.